Quantum CFD and Finite-Element Modeling of a MEG Dewar

A reduced-order cryogenic vessel simulation with quantum probability transport and sensor-waveform coupling

Abstract. Cryogenic magnetoencephalography (MEG) depends on a mechanically stable, thermally isolated dewar. We present a finite reduced-order simulation that couples a quantum-walk-inspired computational fluid dynamics (CFD) surrogate to a thermal-stress model and synthetic MEG source waveforms. The model uses a 20 x 50 flow grid, a 0.13 m inner radius, a 0.15 m outer radius and a 0.4 m vessel height. The equations and plots are intended as a reproducible computational preprint; they do not replace a validated helium property model, structural solver or instrument qualification.

1. Dewar geometry and finite state space

The vessel is represented by cylindrical inner and outer walls. Let theta in [0, 2 pi] be the azimuth and z in [-H, 0] the axial coordinate. For wall radius R in $\{R_i, R_o\}$:

$$x_R(\theta,z) = R\cos(\theta),\quad y_R(\theta,z) = R\sin(\theta),\quad z = z;\quad R_i = 0.13\text{ m},\ R_o = 0.15\text{ m},\ H = 0.40\text{ m}.$$

The CFD state is sampled on N_r x N_{theta} = 20 x 50 = 1000 cells. A complex amplitude $psi_{i,j}$ is initialized uniformly; its squared magnitude is interpreted as a normalized transport weight, $sum_{i,j} |psi_{i,j}|^2 = 1$.

2. Quantum CFD surrogate

The computational fluid model uses a bounded convection pattern to represent a quantum-walk-inspired flow field. With normalized coordinates xi and eta, the velocity surrogate and pressure perturbation are:

$$v(\xi,\eta) = \sin(2\pi\xi)\cos(\pi\eta),\quad \delta p(\xi,\eta) = p_0\sin(10\pi\xi)\cos(10\eta),\quad p_0 = 5000\text{ Pa}.$$

The total pressure is p = 101325 Pa + delta p. A finite update may be written as $psi^{(q+1)} = U_q\, psi^{(q)}$, where $U_q^* U_q = I$ and q = 0,...,Q-1. In this preprint, the displayed field is a deterministic spatial surrogate for the returned simulation snapshot rather than a claim of a full Navier-Stokes or two-fluid helium solution.

3. Thermal stress and strain

For a constrained wall, the leading thermal stress scale uses Young's modulus E, expansion coefficient alpha and temperature drop Delta T = 300 - 4 = 296 K:

$$\sigma_0 = E\,\alpha\,\Delta T,\quad \sigma_v(z) = \sigma_0[1 + 0.5\exp(-10 z\blacksquare)],\quad \epsilon(z) = \sigma_v(z)/E,$$

where z■ is the normalized axial coordinate. The local model uses E = 70 GPa and alpha = 23 x 10^{-6} K^{-1}. The factor in square brackets is a bounded concentration term that emphasizes the cold-end region; it is not a substitute for a mesh-converged finite-element boundary-value problem.

4. MEG source coupling

To connect vessel conditions to a representative MEG signal, the source simulator generates finite alpha and gamma components over t in [0,1] s:

$$a(t) = \sin(2\pi\, 10t)\, w(t),\quad g(t) = \sin[2\pi(30t + 10t^2)]\exp[-(t - 0.5)^2/0.02],$$

where w(t) is a Tukey-like finite taper. A simple sensor observation is $y_m(t) = l_m^T[a(t)q_a + g(t)q_g] + n_m(t)$, with lead vectors q_a, q_g and noise n_m. This separates the cryogenic engineering state from the neural source state while retaining a measurable interface for future co-simulation.

5. Plots and characteristics

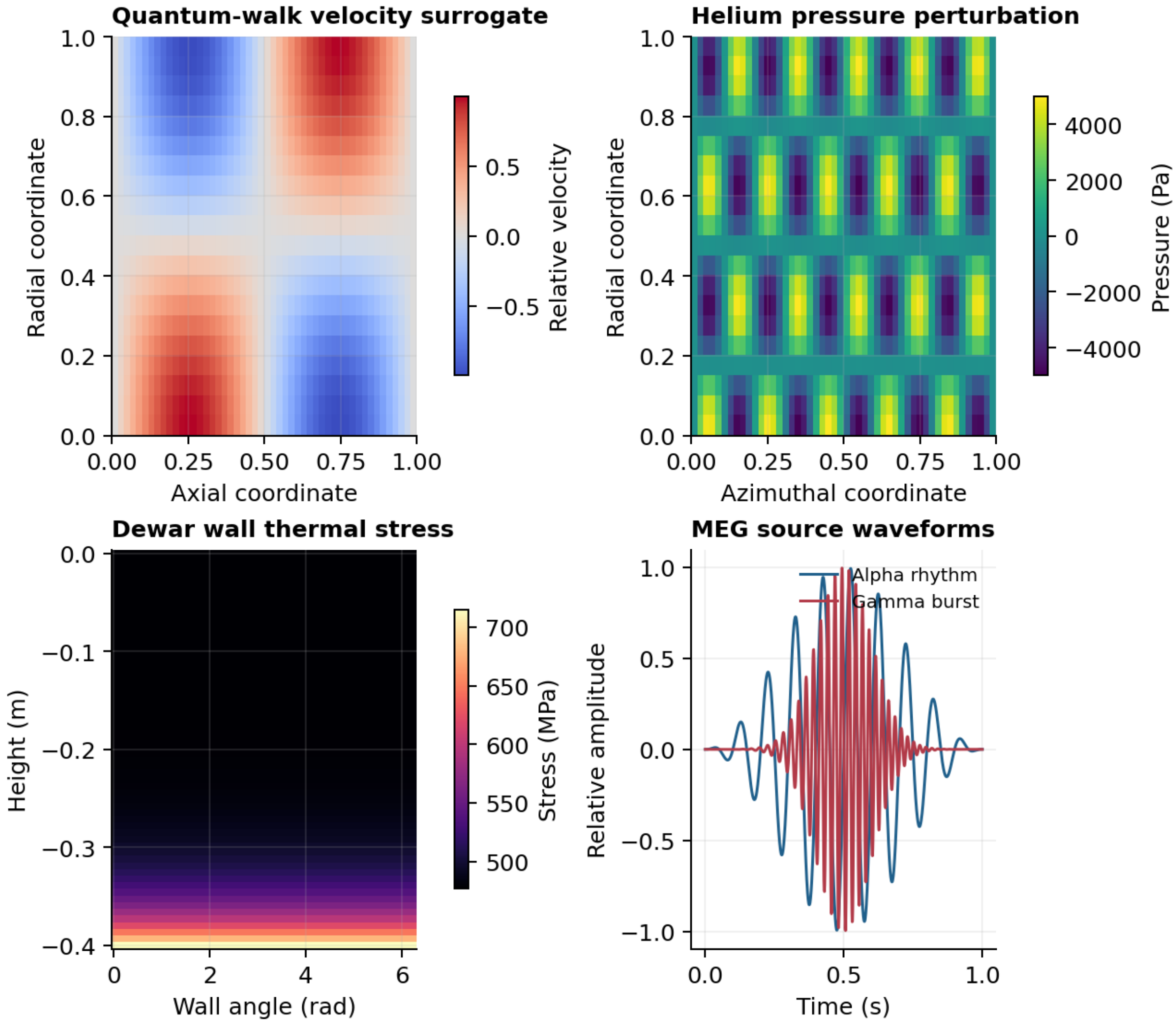

Figure 1. Finite simulation outputs: quantum-walk-inspired velocity, helium pressure perturbation, axial thermal-stress concentration and representative MEG alpha/gamma source waveforms.

Component	Characteristic	Finite setting
Dewar	Cylindrical cryogenic vessel	R_i = 0.13 m; R_o = 0.15 m; H = 0.40 m
CFD	Bounded convection and pressure snapshot	20 x 50 = 1000 cells; p_0 = 5000 Pa
Thermal	Cold-wall stress concentration	4 K to 300 K; E = 70 GPa
MEG	Alpha rhythm and localized gamma burst	10 Hz and 30-50 Hz chirp; 1 s window

6. Limitations and conclusion

This Nature-style preprint reports a transparent reduced-order simulation. Quantitative engineering decisions require temperature-dependent helium thermodynamics, turbulence or two-fluid physics where appropriate, vacuum boundary conditions, material-property curves, contact constraints, mesh convergence and experimental validation. Within those limits, the finite state-space formulation provides a compact test bed for linking dewar pressure/stress characteristics to MEG waveform simulations and later quantum optimization experiments.